Title: Aegis Learn- A Generative AI Personalized Learning Framework

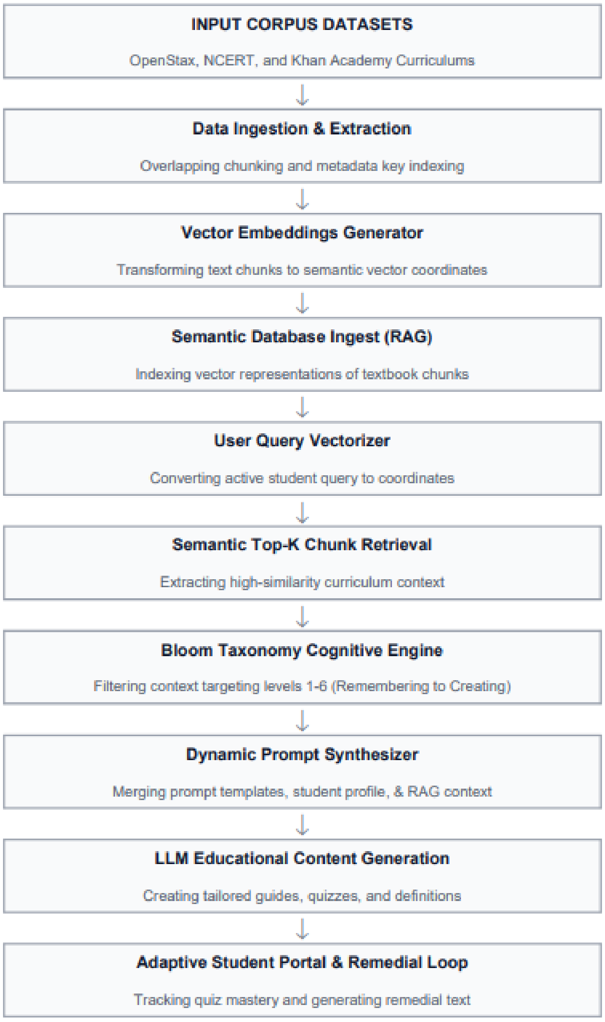
Abstract

Academic textbooks are rich resources of foundational knowledge, but their static nature fails to accommodate the diverse learning speeds, backgrounds, and cognitive needs of individual students. Conventional educational technology often delivers uniform content without dynamic adaptation, leading to student disengagement or unresolved learning gaps. This project proposes Aegis Learn, a Generative AI-powered personalized learning framework that bridges the gap between static textbooks and tailored academic guidance. Aegis Learn integrates Retrieval-Augmented Generation (RAG), custom prompt engineering, and Bloom's Taxonomy to synthesize custom educational materials and drive an adaptive learning cycle. The framework operates by ingesting corpus datasets from open-educational platforms such as OpenStax, NCERT, and Khan Academy, chunking and indexing them for semantic vector matching. During retrieval, student queries dynamically extract relevant textbook context. The system then applies a cognitive filter mapping to the six levels of Bloom's Taxonomy (from Remembering to Creating) to adjust the complexity and cognitive demand of the generated explanations, quizzes, and study guides. Furthermore, Aegis Learn incorporates a student performance loop that automatically evaluates responses, diagnoses concept weaknesses, and triggers remedial content generation to foster mastery. Evaluation of the framework demonstrates its capacity to maintain instructional alignment, suppress hallucinated information by grounding responses in verified textbook contexts, and personalize learning paths. By providing an interactive platform that includes a Bloom Content Generator, Prompt Sandbox, and Student Portal, Aegis Learn showcases the potential of modern LLMs to act as adaptive, transparent, and scalable personal tutors.

Datasets

Dataset	URL link
OpenStax Textbook Corpus	openstax.org
NCERT Textbook Corpus	ncert.nic.in
Computer Science question-answer dataset	https://huggingface.co/datasets/Danhpham2000/computer_science_qa_dataset

Approach



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